



DHEERAJ MARMAT

01 APRIL 2026

Data Cleaning & Preparation

Checks performed:

- Duplicate Order_ID
- Missing traffic delay
- Date validation

Result:

- Verified uniqueness of Order_ID (no duplicates found)
- Checked for missing traffic delays and invalid delivery dates (no issue)
- Validated
 - There is no Actual_Delivery_Date is before Order_Date
 - Standardized date columns (converted TEXT → DATE)

Dataset passed all data quality checks and was standardized for accurate analysis.

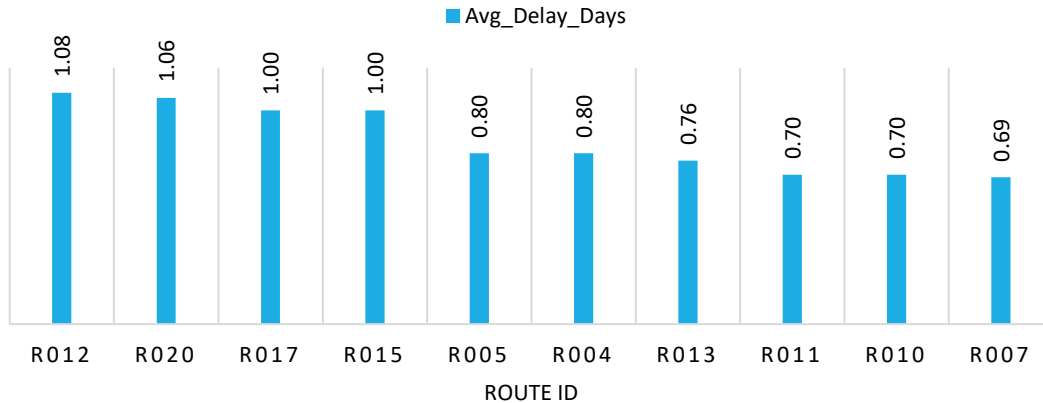
Delivery Delay Analysis

Delay Rate
44%

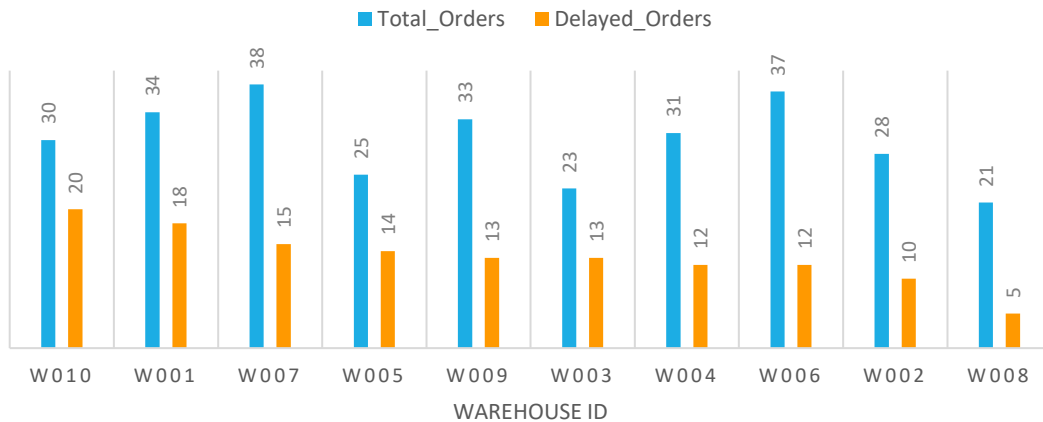
Avg Delay
0.69 days

Worst Route Avg Delay
1.08 days (R012)

AVG DELAY DAYS VS ROUTE



ORDERS VS WAREHOUSE



Insights:

Delays are common:

Even though the average delay is less than 1 day (0.69 days), 44% of orders are delayed, which is a significant impact on customers.

Few routes cause most delays:

Routes R012 and R020 have the highest average delays, making them the main contributors to late deliveries.

Warehouse delays are systemic:

Multiple delayed orders occur within the same warehouses, showing process issues, not one-time problems.

Recommendation:

Focus first on routes R012 and R020 and review warehouse processes to reduce frequent delays and improve on-time delivery.

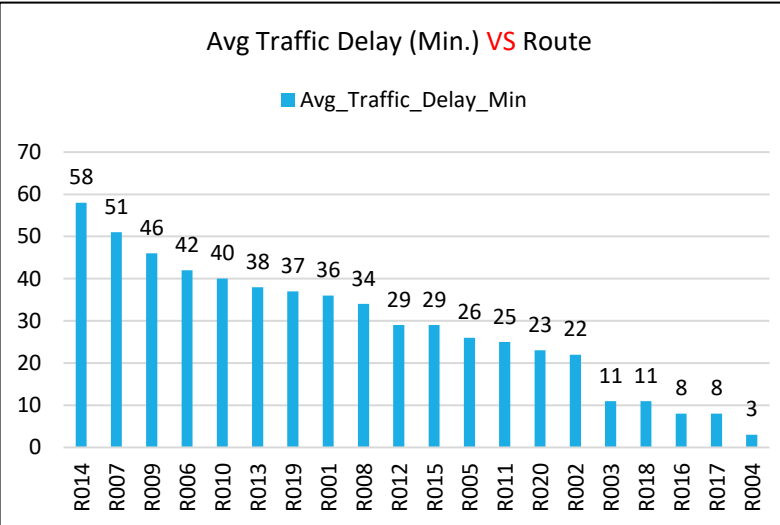
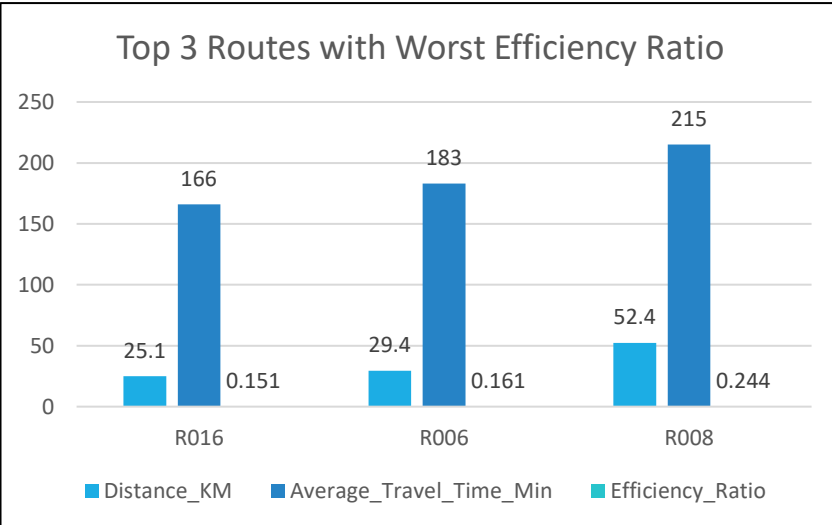
CTRL + Click here to view the query and the corresponding result.

Route Optimization Analysis

Worst Efficiency Routes:
R016 (0.15),
R006 (0.16),
R008 (0.24)

Highest Avg Delivery Time:
R015 (6.33 days),
R017 (6.29 days)

Highest Delay % Route:
R020
(70.6% delayed shipments)



Highest traffic Routes:
R014 (58 min avg),
R007 (51 min avg)

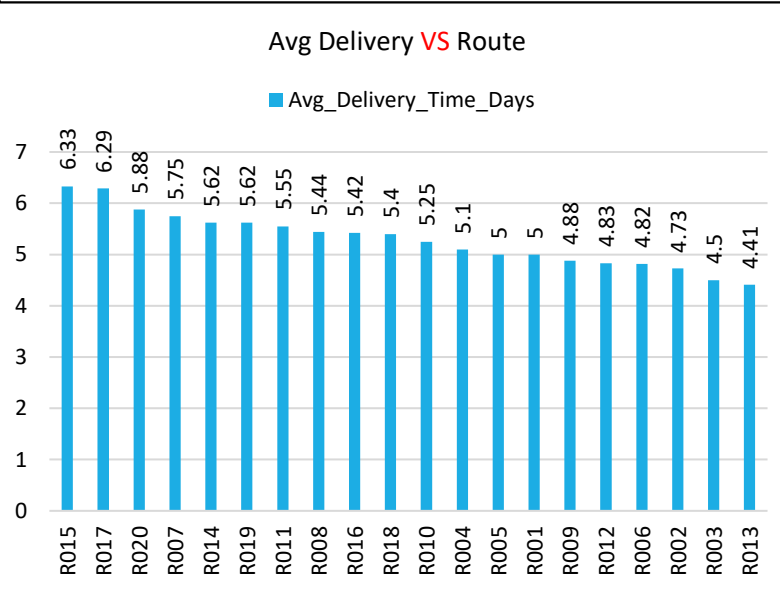
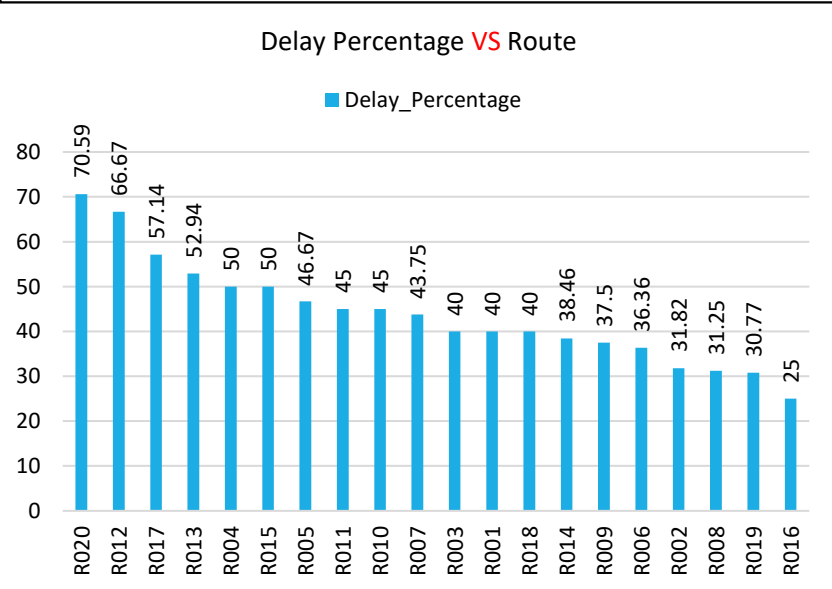
Highest traffic Routes (Avg):
R014 (58 min),
R007 (51 min)

Insights:

- Some routes are highly inefficient:**
Routes R016, R006, and R008 take a long time to cover short distances, increasing cost and inefficiency.
- Long delivery times on specific routes:**
Routes R015 and R017 take the longest time to complete deliveries, increasing customer wait time.
- High delay frequency on many routes:**
Routes like R020 and R012 have more than 60% delayed shipments, showing serious reliability issues.

Recommendation:

Prioritize optimization of routes R016, R006, and R008 for efficiency, and address high-delay routes like R020 to improve delivery reliability.



Warehouse Performance Analysis

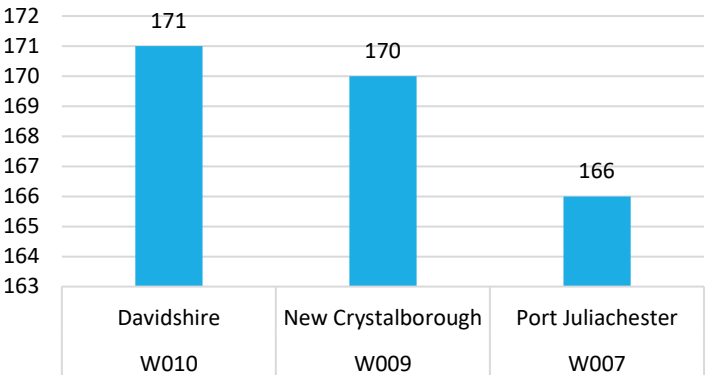
Slowest Warehouses
(Processing Time)
W010 – 171 mins
W009 – 170 mins
W007 – 166 mins

Bottleneck Warehouses
(Above Avg Processing Time)
W010, W009, W007, W008, W001

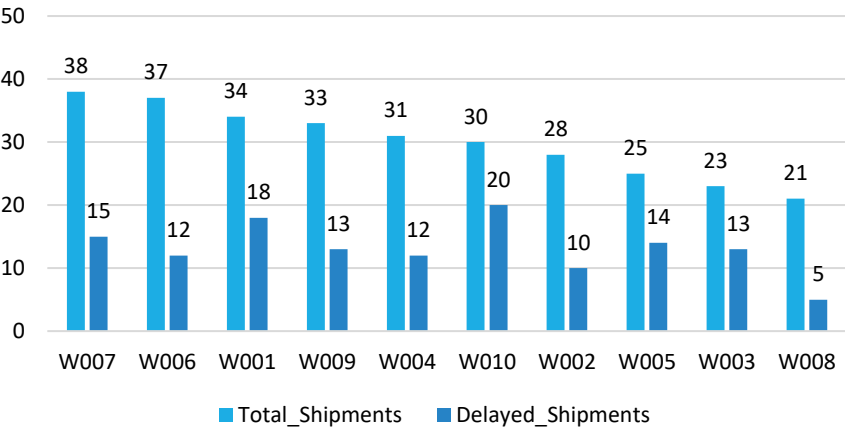
Lowest On-Time Delivery
W010 – 33.3% on-time

Top 3 Warehouse with highest Processing Time

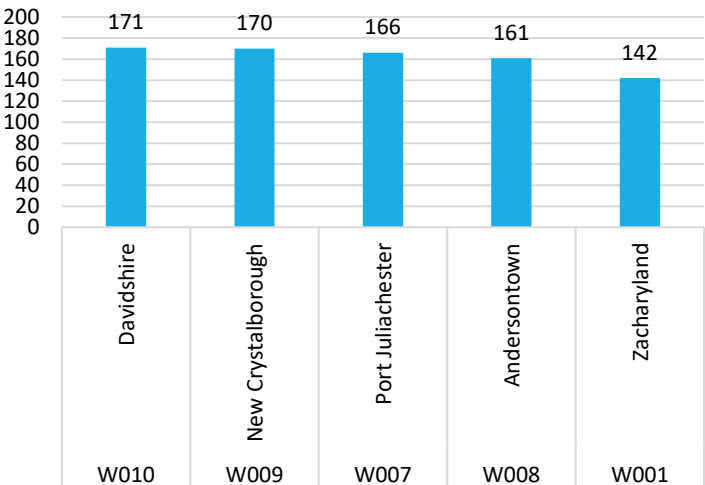
Processing_Time_Min



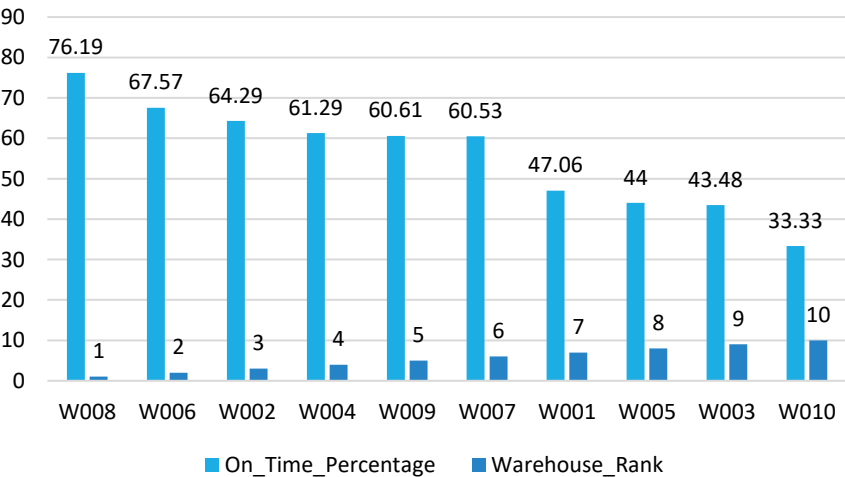
Total VS Delayed Shipment Per Warehouse



Bottleneck WH (Processing time > Global avg)



On Time Delivery % VS Warehouse



Insights:

- Some warehouses are slow:**
Warehouses W010, W009, and W007 take the longest time to process shipments, slowing down deliveries.
- Delays are uneven across warehouses:**
Warehouses like W010 and W001 have a high number of delayed shipments compared to others.
- Performance varies significantly:**
While W008 performs best with the highest on-time delivery rate, W010 performs worst, delivering on time only 33% of the time.

Recommendation:

Prioritize process improvement in warehouses W010, W009, and W007 to reduce processing delays and improve overall on-time delivery performance.

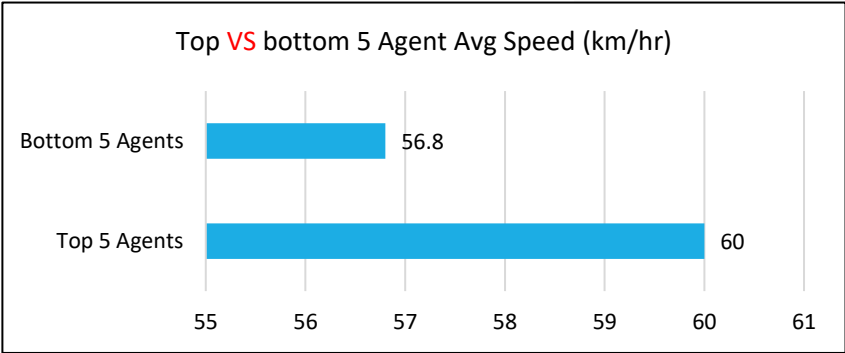
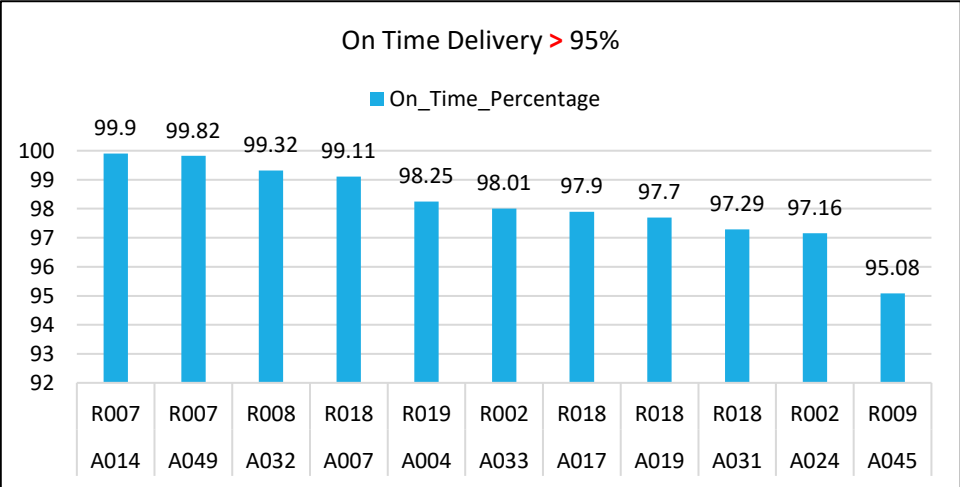
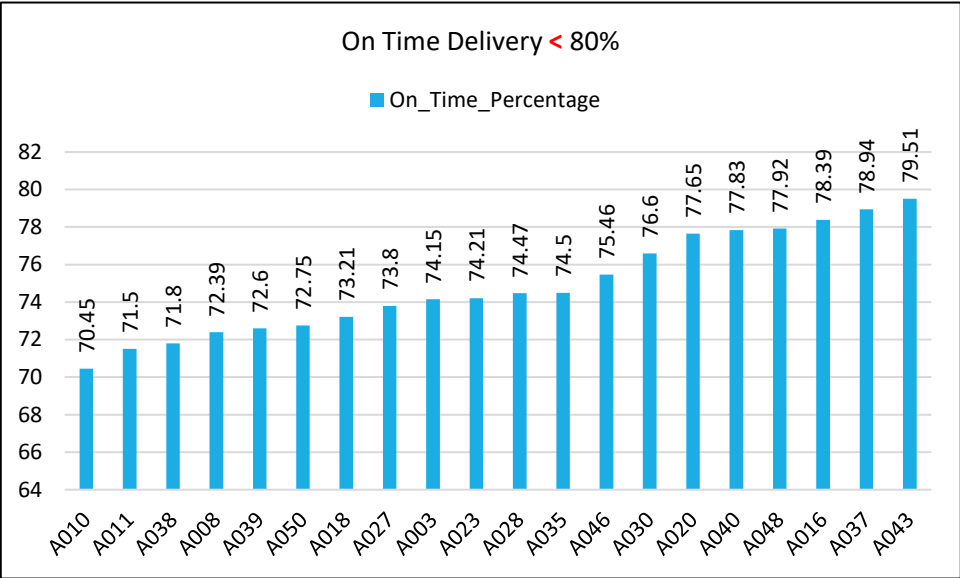
Delivery Agent Performance Analysis

Total Agents
50

Avg Speed
58.78 km/hr.

Total Agents whose
(Speed > Avg Speed) - 27
(Speed < Avg Speed) - 23

Best Agent Performance
(>99% on-time delivery)
A014, A049, A032, A007



Low Performing Agents
20 agents have on-time delivery below 80%

Speed Comparison
Top 5 Agents Avg Speed: 60 km/hr.
Bottom 5 Agents Avg Speed: 56.8 km/hr.

Insights:

- Agent performance varies by route:**
Some agents consistently perform better than others on the same routes.
- Several agents underperform:**
Around 20 agents have on-time delivery below 80%, indicating a need for performance improvement.
- Speed alone is not enough:**
Top agents are only slightly faster than bottom agents, showing that reliability matters more than speed.

Recommendation:

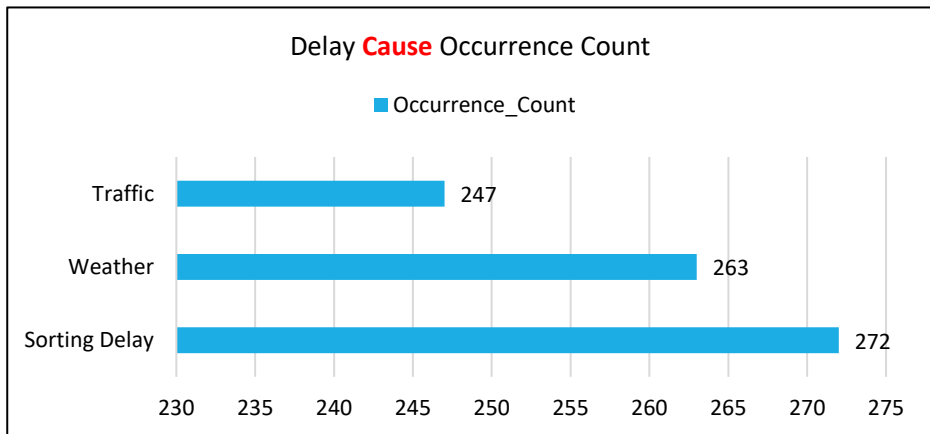
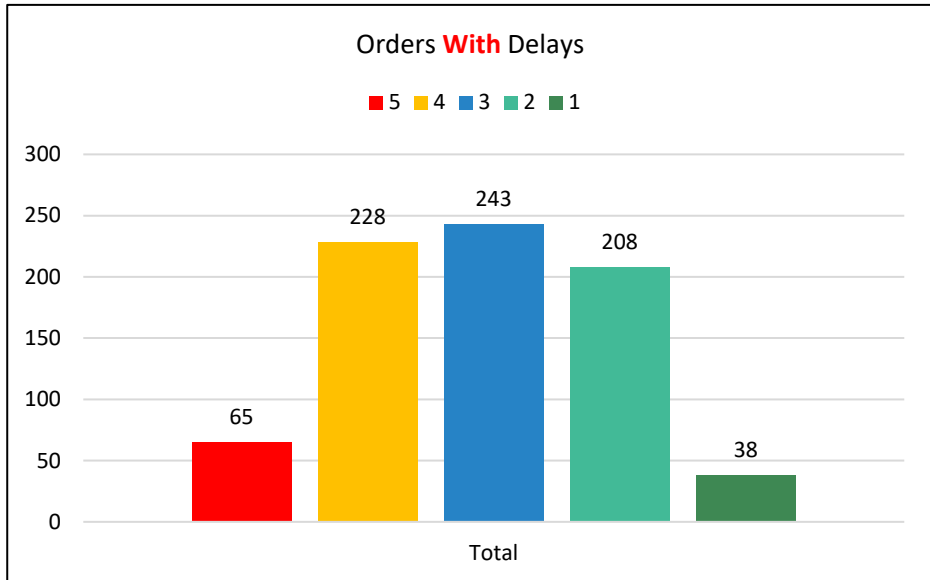
Provide training and monitoring for agents with on-time delivery below 80%, and use best-performing agents as benchmarks to improve overall delivery reliability.

Shipment Tracking Analytics

Total Shipment: 1040
Total Delay Orders: 782
Total No Delay Orders: 258

Delay causes:
Sorting: 272 (34.8% Highest)
Weather: 263 (33.6%)
Traffic: 247 (31.6%)

High risk orders:
65 orders has 5 delays
228 orders has 4 delays



Insights:

- **Sorting delay is the biggest issue:**

It is the #1 cause of checkpoint delays, meaning internal hub/sorting operations need attention.

- **Delay are caused by multiple sources:**

Weather and traffic are almost as frequent, so delays aren't caused by only one factor.

- **Some orders face repeated disruption:**

Several orders have 4–5 delayed checkpoints, making them high-risk shipments that need monitoring.`

Recommendation:

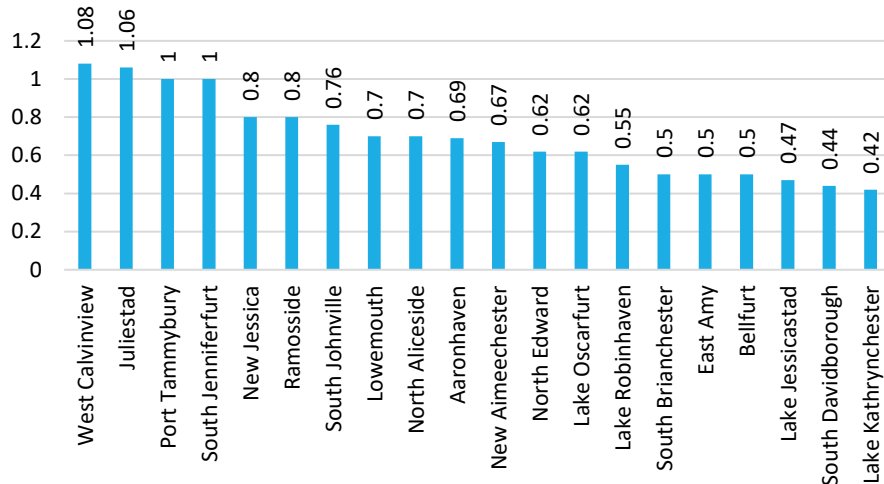
Prioritize reducing sorting delays at hubs and actively monitor high-risk orders (4–5 delayed checkpoints) for faster escalation and rerouting.

CTRL + Click here to view the query and the corresponding result.

KPI Summary – Delivery Performance

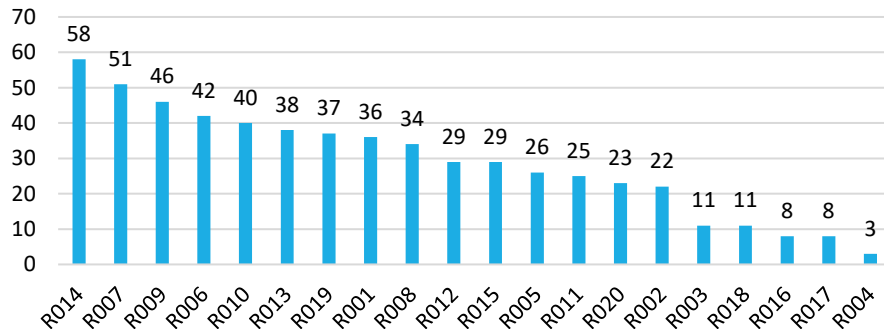
Region **VS** Avg Delivery Delay Days

Avg_Delivery_Delay_Days



Avg Traffic Delay (Min.) **VS** Route

Avg_Traffic_Delay_Min



Overall Delivery

56% On-Time Delivery
44% Deliveries Delayed

Highest Traffic Delay Routes

R014 – 58 min
R007 – 51 min
R009 – 46 min

Highest Avg Delay Regions

West Calvinview – 1.08 days
Juliestad – 1.06 days

Worst region avg delay = 1.08
vs best region delay = 0.42
(gap 0.66 days)

Insights:

Overall performance needs improvement:

Overall delivery performance is below target, with only 56% on-time deliveries.

Delays vary by region:

Regions like West Calvinview and Juliestad experience the highest average delivery delays.

Traffic congestion is a major factor:

Routes R014, R007, and R009 face the highest traffic delays, increasing delivery risk.

Recommendation:

Focus on improving delivery performance in high-delay regions and optimize traffic-heavy routes to increase the overall on-time delivery rate.

Video Summary :

<https://www.loom.com/share/9fd9a4c278d7485586d100c4efc4a8bf>

CTRL + Click here to view the query and the corresponding result.



Thank You